MIL-S-45108C(MU)
31 January 1964
SUPERSEDING
MIL-S-15108B(ORD)
10 AUGUST 1962

MILITARY SPECIFICATION

SIGHTS, INFINITY M44 and M44C

1. SCOPE

1.1 This specification covers two similar sights known as Sight Infinity, M44 and M44C (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or *requests* for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830 — Optical Components for Fire Control Instruments; General Specification Governing the Manufacture and Inspection of.

ML-F-13926 — Fire Control Materiel; General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel;; Packaging of.

MIL-I-45208 -Inspection Requirements, General Specification for.

STNDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes.

MIIATD-109 -Quality Assurance Terms and Definitions

DRAWINGS - -

U. S. ARMY MUNITIONS COMMAND

F8229017 — Sigh, Infinity, M44.

F8598949 — Sight Infinity, M44C.

LSQAP-8229017 — List of Supplementary Quality Assurance Provisions for Sight, Infinity, M44.

FSC 1240

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LsQAP - 8598949 . - List of Supplementary Quality Assurance Provisions for Sight, Infinity, M44C.

IEL-8229017 — Index of Inspection Equipment List for Sight, Infinity, M44.

IEL-8598949 — Index of Inspection Equipment List for Sight, Infinity, M44C.

PACKAGING DATA SHEET

MIL-P-14232/ P8229017 — Packaging of Sight, Infinity, M44 and M44C.

(Copies of specification, standards, drawingsdegrees and publications required by supplier's in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting other.)

3. REQUIREMENTS

3.1MaterIALS. The materials shall be in accordance with the drawings, material specifications, and general specifications forming a PART of this specification.

3.2 Fabrication. The Sight shall be manufactured in accordance with the drawing applicable to the model of the sight being procured and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specificationS. The s i g h t shall meet the following requirements of Specification MIL-O-13830 and MILF-13926.

3.3.1 MIL-O-13830.

(a) Cleanliness

(b) Fixed eyepiece focus

(c) Illuminated reticles

3.302 MIL-F-13926.

-- (a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Part identification and marking.

(e) Workmanship.

3.4 Alinement With the sight positioned as shown on Drawing F8289200 for Sight M44 and Drawing F8598951 for Sight. M44C with the mounting surface 20 degrees plus or minus 1 minute from the vertical plane and the key horizontal within plus or minus 1 minute, the line of sight from the center of the reticle circle shall be perpendicular to the vertical plane as adjusted by the boresight knobs

3.4.1 Knob travel With the geometric axis as positioned in 3.4, the boresight knobs shall provide at least 10 roils movement in both directions (total excursion of 20 mils) for azimuth and for elevation.

3.5 Environmental.

$5.1 Storage temperature. The sight shall show no evidence of physical failure and shall meet all performance requirement of this specification at a standard ambient temperature of 60° to 90°F. after having been exposed to and thermally stabilized at ambient temperatures of -80° and + 160° F.

3.5.2 Operating temperatures. The sight shall meet the appliable requirements of 3.7 while exposed to and thermally stabilized

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at ambient temperatures of -40° and + 150°F.

3.5.3 Vibration With the sight mounted in its normal operating position it shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 6 minutes plus or minus 15 seconds.

3.6 Performance

3.6.1 Reticle retention. After being subjected to the requirements of 3.5.3, the line of sight shall not have changed by more. than 0.5 mil from the setting established and recorded in 3.4.

3.6.2Focus. The reticle ring target shalll be focused between 500-1000 meters.

3.6.3 Reticle. . The inner diameter of the reticle circle shall subtend an angle of 21 mils ± 3 mils.

3.6.4 Sealing. The Body Ass e m b 1 Y D8229018 of Sight M44, or the Body Assembly D8598950 of Sight M44C shall show no evidence of leakage when subjected to an internal pressure of 5 psi (gage pressure) for a period of five minutes.

3.6.5 Illuminatim. Utilizing a 24 volt source to illuminate the reticle, the reticle pattern shall appear uniformly illuminated and free of ragged edges or breaks throughout its entire circumference.

3.7 Operability.

3.7.1 Boesight knobs. The torque required to rotate the azimuth and elevation boresight knobs shall not exceed 7 pound inches at room temperature (plus 60 to 90 degrees Fahrenheit) and at the operating temperature specified in 3.5.2.

3.7.2 Rheostat. The torque required to rotate the rheostat knob shall not exceed 4 pound inches, and rotation in both directions shall cause illumination of the reticle to vary appropriately when checked at the standard ambient temperature (Plus 60°to 90°F) and the operating temperatures specified in 3.52.

4. QUALITY ASSURANCE PROVISIONS

U Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government Inspection records of of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections Set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein from the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL-I-45208 and the component inspection requirements of Specification MIL-F-13926 Definitions of inspection terms not other defined herein shall be as listed in Standard MIL-STD-109.

42 Inspection provisions

4.2.1 Inspection lot. Unless otherwise specified by the contracting officer, inspection lot size, formation and presentation of lots shall be in accordance with “Submission of Product” and Drawing of Samples” specified in Standard MIL-STD-105.

4.2.2 Examination and tests. Components and subassemblies shall be inspected

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in accordance with the inspection provisions contained in Specification MIL-F-13926 and List of Supplementary Quality Assurance Provisions, LSQAP-8229017 for Infinity Sight: M44 and LSQAP-8598949 for ln-finity Sight: M44C. Examinations and tests related to Section S herein shall be per-formed on a defect (individual character-istic) basis in accordance with Standard MIL-STD-105, and the inspection level sampling plans specified in Table I "Classification of Defects". Examinations and tests for Packaging, Packing and Mark-ing shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall con-stitute the minimum inspection to b.e per-formed by the supplier prior to Government acceptance or rejection by lot. The Govern-ment reserves the right to inspect for any applicable requirement and and to reject in-dividual non-conformning items.

Table I. Classification Of Defects

Use General Inspection level11 of Table I, with

Critical: None

Major: AQL 0.65 percent defective

101. Vibration	**3.5.3**	**4.5.5**
102. Alinement	**3.4**	**4.5.1**
103. Knob travel	**3.4.1**	**4.5.2**
104. Reticle retention	**3.6.1**	**4.5.6**
105. Focus of reticle	**3.6.2**	**4.5.7**
106. Reticle	**3.6.3**	**4.5.8**
107. Illumination of reticle	**3.6.5**	**4.5.10**
108. Boresight knobs operability	**3.7.1**	**4.5.11**
109. Rheostat operability	**3.7.2**	**4.5.12**

Minor: AQL 2.5 percent defective

201. Cleanliness	**3.3.1(a)**	**Visual—MIL-O-13830**
202. Identification and marking	**3.3.2(d)**	**Visual—MIL-F-13926**

Note: The above tests shall be performed at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit

4.2.3 Disposition of non- conforming product. Rejected lots shall be screened for all defective characteristics. Removal or cor-rection of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

493 Control tests.

4.3.1 General. One sight of each model shall be selected at random from each 100 produced, or from each months production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the fol-lowing examinations and tests:

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Control test	Requirement	Test procedure
(a) Material.	3.1	kVisual-specifications Dwgs
(b) Fabrication.....	3.2	Visual-Standard Measuring Equipment
(c) General specification	3.3	MIL-O-13830- MIL-F-13926

4.3.2 Environmental Three sights of each model shall be selected at random from each 50 produced or from each months production, whichever Occurs first Control samples shall have successfully met all acceptance teats specified herein prior to conducting the following examination an

Control test	*Requirement*	*Test procedure*
(a) Storage temperatures	**3.5.1**	**4.5.2**
(b) Boresight knobs operability at extreme temperatures	**3.7.1**	**4.5.11**
(c) Rheostat operability at extreme temperatures	**3.7.2**	**4.5.12**
(d) Sealing	**3.6.4**	**4.5.9**

4.%3 COntrol test *failure.* Should a control test sample of either model sight fail to meet the specified test requirements acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted Samples have been approved.

4.4 Inspection equipment Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIXA-45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in *accordance* with "Inspection Equipment", and "Government Furnished Material" requirements of Specification MIL-I-45208.

4.4.2 Contractor *furnished inspection equipment.*

*4.4.2.1Government design. An inspec*tion equipment required by drawings forming a part of the contract and not provided by the Government shaIl be supplied by the contractor in accordance with the List of lnspection Equipment Numbers, IEL-8229016 for Infinity Sight: M44 and IEL-8598949 for Infinity Sight: M44C. Alternate designs may be SUbstituted only as provided by "Inspection Provisions" of Specification MIL-1-45208.

4.4.2.2 Contarctor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with

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the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shaH be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 Equipment to test for environmental requirements. All environmental testing equipment used to determine conformance with the temperature and vibration requirements shall conform to the "Test Facilities" requirements contained in Specification MIL-F-13926.

4.4.2.4 Equipment to test for performance requirements. The special testing equipment shall position the infinity sight on a keyed locating surface with the mounting surface 20 degrees plus or minus 1 minute from the vertical plane and the key horizontal. The test equipment shall be equipped with a medium capable of measuring the projected reticle image of the infinity sight, in increments of 1 mil within an accuracy of plus or minus 0.2 mil in both horizontal and vertical directions, in a manner suitable for measuring the angular subtense of the projected reticle pattern, and boresight adjusting requirements. The equipment shall 'be capable of relocating any infinity sight within 0.05 miI of the original mounting position. Provisions shall be made to adjust ,the measuring medium with respect to the test equipment mounting surface and the horizontal locating key to an angular accuracy of 1 minute respectively,

4.4.2.5 Boresight knob. The equipment shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to ten pound inches, the accuracy of which shall be within plus or minus one-quarter pound inch throughout the entire zone of excursion. It shall be equipped with an adapter to accommodate the boresight knob.

4.4.2.6 Rheostat knob. This equipment shall consist of a torque wrench type tester which meets the requirements specified in 4.4.2.5 and shall be equipped with an adapter to accommodate the rheostat knob.

4.4.0.7 Equipment to test sealing. The equipment to test sealing shall be capable of charging the interior of the infinity sight to an internal pressure of at least 5 psig of nitrogen having a dew point of at least as low as minus 25 degrees Fahrenheit. The equipment shall be fitted with a pressure gage capable of recording the internal pressures up to at least 5 psig, sub-divided into increments of .1 psig. The accuracy of the pressure gage shall be within one percent at any point on the scale.

4.5 Test methods and procedures.

4.5.1 Alinement. The infinity sight shall be positioned and alined using special test equipment as defined in 4.4.2.4 to hold the sight as specified in 3.4. An auxiliary telescope (projected collimator) with an attached eyepiece will aid in measuring the projected reticle image of the infinity sight. The collimator used shall contain a reticle with either a mil grid pattern or cross lines graduated in roils spaced about the geometric axis of the collimator target for each model of sight. Connect a 24 volt power supply to the contact plug of the sight. Rotate variable resistor knob cIockwise to illuminate reticle. The line of sight through the center of the illuminated reticle shall be directed to the reticle of the collimator, using the boresight knobs to center the image on the mil grid of the collimator.

4.5.2 Knob travel. Conformance with the requirements of 3.4.1 will be determined by visual and manual examination, with [he

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sight positioned as in 4.5.1 and the reticle in focus in the central position. Observe upper edge of the illuminated reticle. Rotate elevation boresight knob counterclockwise until the upper edge of the sight reticle reads at least 10 mils in movement from its central position. Return circle to center position by clockwise rotation of knob. While observing the lower edge of the illuminated circle and clockwise rotation of knob, bring the circle down until the lower edge reads at least 10 mils in movement. Return reticle to central position. Repeat this procedure with the azimuth boresight knob. Range of travel in azimuth shall move the reticle image at least10 mils right and 10 mils left of center.

4.5.3 Storage temperatures. The sight shall be exposed to the temperatures indicated in 3.5.1 utilizing test facilities conforming with the requirements of 4.4.2.3. The Tkemperatures shall be changed gradually to avoid thermal shock After the sight has been exposed to the extreme temperatures and permitted to return to standard ambient temperature (60° to 90°F) the sight shall be visually examined for defects and shall subsequently be inspected for all performance requirements of this specification.

45.4 Operating temperatures. The sight shall be exposed to the temperatures indicated in 3.5.2 employing the same test facilities as used for the temperature test. of 4.5.3. The temperatures shall be changed gradually to avoid thermal shock While stabilized at each extreme temperature condition the sight shall be tested in accordance with 4.5.11 and 4.5.12

4.5.5 Vibration. After inspection for boresight knob travel, as outlined in 4.5.2, care shall be exercised not to disturb the settings of the boresight knobs when removing the sight from the special test equipment. Each infinity sight in the sample shall be secured to the vibrating table in the position specified in 3.5.3 by means of a special holding medium and vibrated to the requirements of 3.5.3. At the conclusion of this vibration remove the sight from the vibration tester and subject the sight to a visual and tactile examination for evidence of damage then remount the sight on the special test equipment as During this_ phase, care shall be exercised not to disturb the boresight knob setting.

45.6 Reticle retention Subsequent to the test 0f 4.5.5 the line of sight shall be checked again in accordance with 4.5.1. the position of the line of sight shall be identical to the original setting in 4.5.1. Any difference in these resulting from the vibration test, shall not exceed the value specified in 3.6.1.

4.5.7 Focus. The objective of the anxiliary telescope (collimater)shall be set between 500 and 1000 meters, and the collimagrad- tor eyepiece adjusted Centralize the reticle image of the sight unit under test by ment of the boresight knobs. Observe the image of the eight reticle formed on the collimator reticle The sight reticle image must be in sharp focus on the collimator reticle to comply with the requirements specified in 3.6.2.

4.5.8 Reticle. The test for the & value of the reticle target ring is accomplished by checking it aginst the collimator containing a reticle target with dimensions suitable for the target distance utilized, and based on the reticle ring dimensions. shown on the drawing applicable to the model of sight under test The error for this test shall not exceed the limit specified in 3.6.3.

4.5.9 Sealing. The sight under test shall be subjected to an internal pressure as indicated in 3.6.4, using dry nitrogen. The test shall be conducted by means of standard pressure testing equipment. Pressure shall be applied by removal of the plug in the body provided for this purpose. During the time period specified, the interior pressure shall not decrease. The interior pressure

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shall be measured and monitored with measuring equipment that meets the requirements of 4.4.2.7. Upon completion of the sealing test, the sight shall then be sealed at atmospheric pressure.

4.5.10 Illumination. Illuminate the reticle lamp by applying the voltage specified in 3.6.5 through a connection connected to the contact plug on the housing assembly. The reticle pattern shall be visually observed to determine compliance with the requirements of 3.6.5 Clockwise rotation of the rheostat knob shall cause the illumination of the reticle to get brighter and counterclockwise rotation shall dim the reticle Illumination.

4.5.11 Boresight knob operability. The running torque required to rotate the specified knobs shall be measured using test equipment that meets "the requirements of 4.4.2.5 Torque shaII be measured while rotating the knobs through its full range of movement. Torque values obstained at minius 40 degrees and plus 150 degrees Fahrenheit shall be in conformance with 3.7.1. -

4.5.12 Rheostat operability. The running torque required to rotate the rheostat knob shall be measured using test equip ment that meets the requirements of 4.4.2.6. Torque shall be measured while rotating the knob through its range of movement in both directions, and illumination of the reticle shall be visually observed to determine compliance with the requirements specified in 3.7.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8229017. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use Infinity Sight, M44 is used in conjunction with Periscope, M29; Infinity Sight M44C is used in conjunction with Periscope, M31 to superimpose an illuminated reticle circle upon the target as seen through the periscope unity power optical system and serves as a sighting reference for the coaxial machine gun.

6.2 Ordering data Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing.

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

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